

PET 212E – Homework 5
Core Analysis: Porosity and Permeability Data
ITU-PDGM Well-1, Ambient Conditions

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Contents

1 Introduction

This report presents the core analysis performed on the ITU-PDGM Well-1 dataset for PET 212E Homework 5. The dataset contains petrophysical measurements obtained under ambient conditions and at 800 psi confining pressure, including porosity (both helium and conventional), grain density, air permeability (K_{air}), and Klinkenberg-corrected permeability (K_{∞}).

The objectives of this assignment are:

1. Plot atmospheric porosity versus grain density and examine the grain density distribution.
2. Plot 800 psi air permeability versus 800 psi helium porosity on a semi-log scale to observe the permeability–porosity relationship.
3. Compare 800 psi helium porosity with atmospheric porosity, including a trendline and equality line.
4. Compare Klinkenberg-corrected permeability (K_{∞}) with air permeability (K_{air}) on a log–log scale and quantify the Klinkenberg effect.

2 Dataset Overview

The dataset covers 43 core plug samples from a single well. Measurements were taken at ambient conditions (temperature $\approx 25^{\circ}\text{C}$, atmospheric pressure) and under 800 psi confining pressure. Key columns used in this analysis are summarised in Table ??.

Table 1: Key petrophysical parameters used in this study.

Parameter	Description
ϕ_{atm}	Atmospheric porosity (%)
ρ_g	Grain density (g/cc)
ϕ_{He}	Helium porosity at 800 psi (%)
K_{air}	Air permeability at 800 psi (mD)
K_{∞}	Klinkenberg-corrected permeability at 800 psi (mD)

3 Results and Discussion

3.1 Plot A: Atmospheric Porosity vs. Grain Density

Figure ?? shows the scatter plot of atmospheric porosity against grain density. A tight cluster around $\rho_g \approx 2.65$ g/cc is consistent with a predominantly quartz-rich (sandstone) lithology. Samples that deviate significantly in grain density may indicate the presence of heavy minerals or authigenic cements.

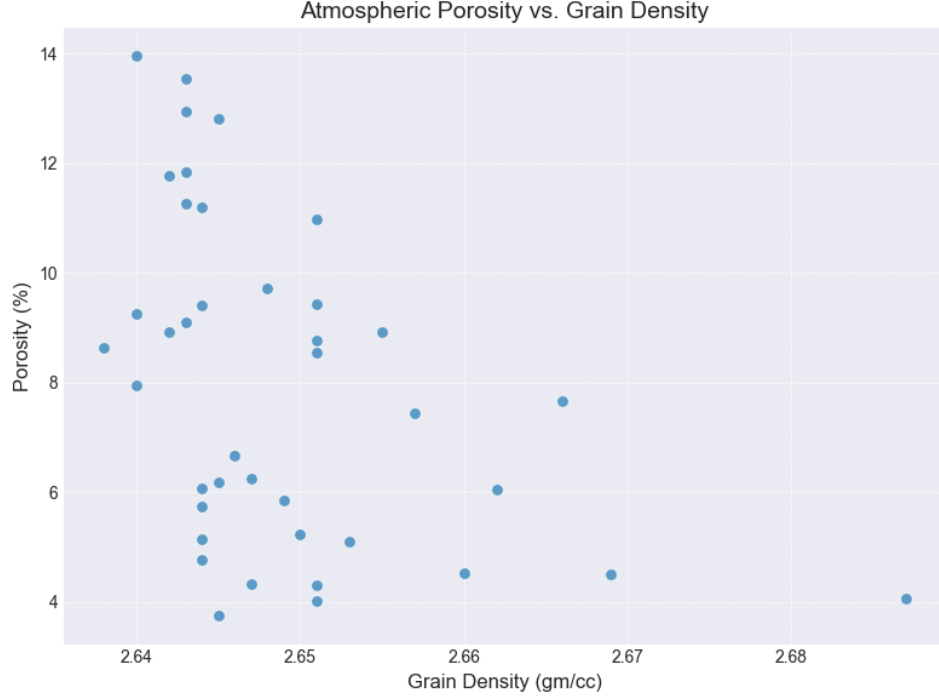


Figure 1: Atmospheric porosity vs. grain density.

3.2 Plot B: 800 psi Air Permeability vs. 800 psi Helium Porosity (Semi-log)

Figure ?? presents the permeability–porosity cross-plot on a semi-logarithmic scale. The positive correlation confirms the expected trend: higher porosity samples generally exhibit higher permeability. The scatter is typical of a heterogeneous reservoir where pore-throat geometry varies independently of total pore volume.

3.3 Plot C: 800 psi Helium Porosity vs. Atmospheric Porosity

Figure ?? compares helium porosity measured at 800 psi confining pressure with atmospheric porosity. The trendline slope is slightly less than 1 (the equality line), indicating that net overburden stress causes measurable pore compaction, reducing effective porosity compared to the unstressed state. The coefficient of determination (R^2) quantifies how well 800 psi helium porosity can be predicted from atmospheric measurements.

3.4 Plot D: K_{∞} vs. K_{air} (Log–Log)

Figure ?? compares Klinkenberg-corrected permeability (K_{∞}) with measured air permeability (K_{air}) on a log-log scale. Points plotting below the equality line ($K_{\infty} < K_{air}$) indicate the Klinkenberg effect: gas slippage at low pressure artificially inflates the apparent permeability of tight samples. The power-law trendline quantifies this systematic bias across the permeability range.

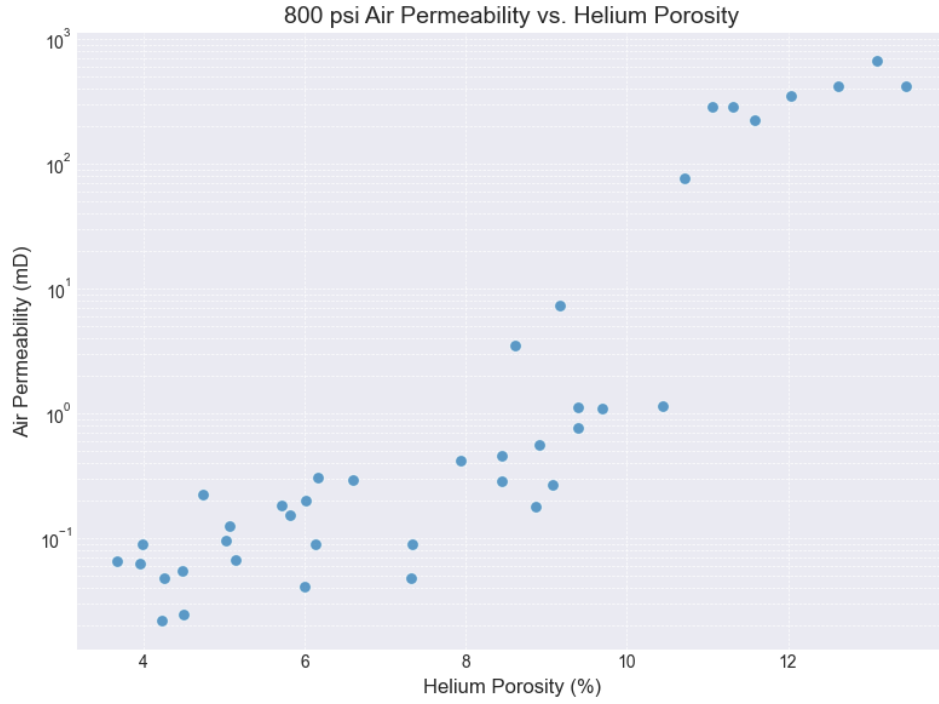


Figure 2: 800 psi air permeability vs. 800 psi helium porosity (semi-log scale).

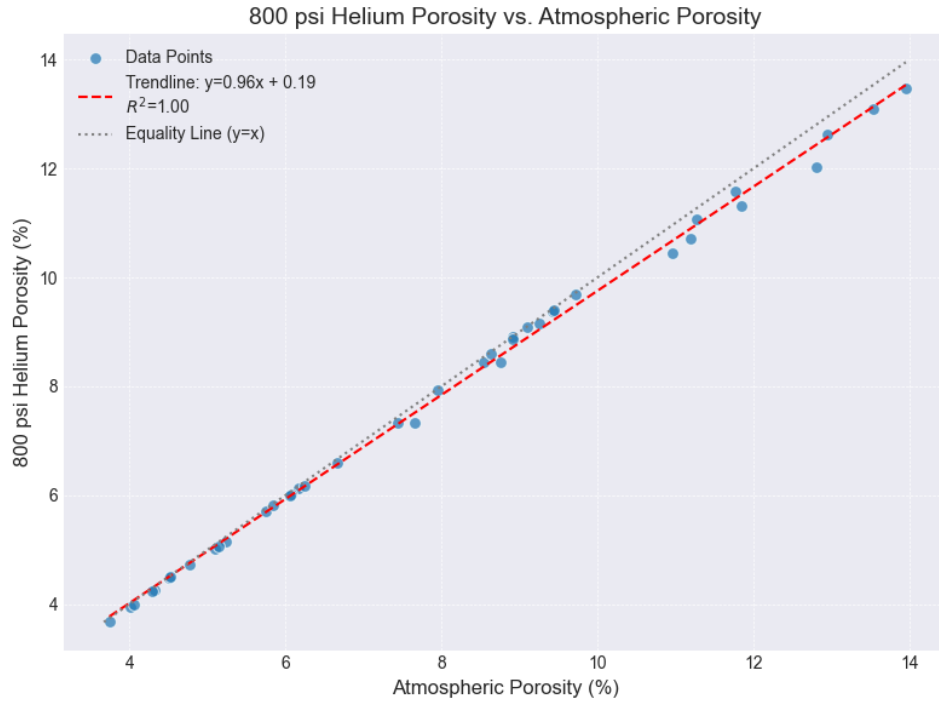


Figure 3: 800 psi helium porosity vs. atmospheric porosity with trendline and equality line.

4 Conclusion

The core analysis of ITU-PDGM Well-1 data confirms:

- The formation is dominated by quartz-rich mineralogy ($\rho_g \approx 2.65$ g/cc).

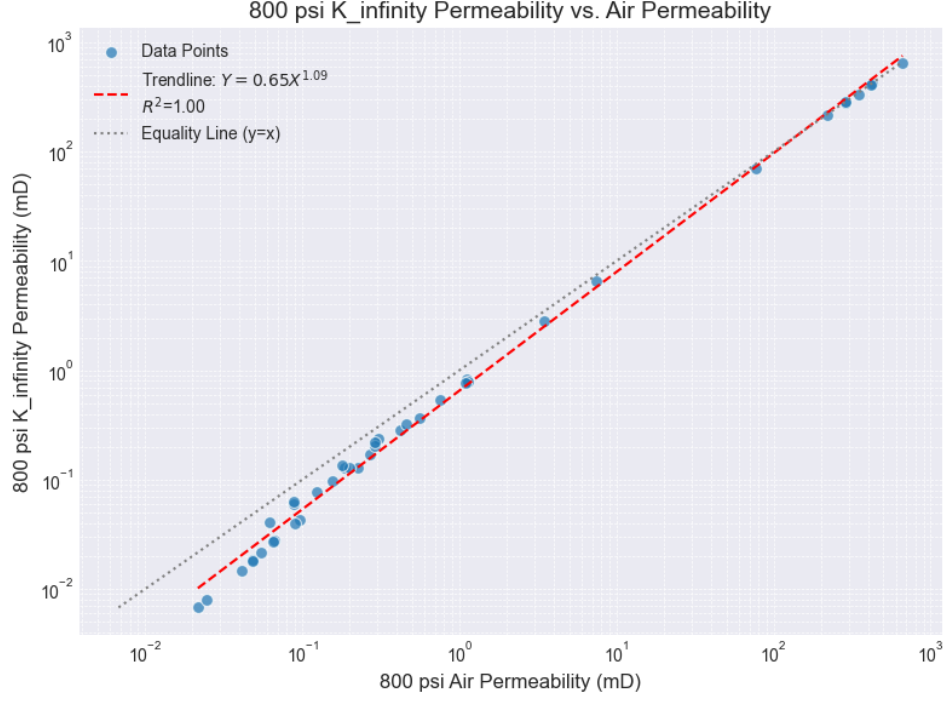


Figure 4: Klinkenberg-corrected permeability (K_{∞}) vs. air permeability (K_{air}) on a log–log scale.

- A positive permeability–porosity correlation exists, though pore-geometry heterogeneity introduces scatter.
- Confining stress (800 psi) reduces porosity compared to atmospheric measurements, as expected from pore compaction.
- The Klinkenberg effect is significant in low-permeability samples, where K_{air} overestimates K_{∞} by a measurable margin.